CO1 ASSESSMENT REPORT

Student Name: Mohammed Mudhassir A

Reg.No: 192424367

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AGRI VISION: CROP-DISEASE SCREENING

1. PRELIMINARY DATASET PREPARATION AND VERIFICATION

1.1 Objective

The objective of this stage was to load a public leaf-disease image dataset, inspect its class distribution, convert the images into a consistent numerical representation, and verify that the prepared data could be loaded successfully in Python using VS Code.

1.2 Dataset Source and Initial Class Selection

The Plant Village leaf-disease dataset was accessed through a Kaggle Notebook. For the initial pilot experiment, three tomato-leaf classes were selected: Healthy, Early Blight, and Late Blight. The grayscale image folders were used as the source for preprocessing.

1.3 Image Preprocessing

Each selected image was converted to grayscale and resized to 64 × 64 pixels. Each image was then flattened into a one-dimensional feature vector containing 4,096 pixel values. Pixel values were stored as unsigned 8-bit integers (uint8), ranging from 0 to 255. The image matrix and corresponding class labels were saved in a compressed NumPy file named agrivision_tomato.npz.

The resulting feature matrix had shape (4500, 4096), while the label vector had shape (4500,). This means that the pilot dataset contained 4,500 images, with 4,096 features per image and one label per image.

1.4 Initial Class Distribution

Healthy: 1,591 images

Early Blight: 1,000 images

Late Blight: 1,909 images

Total: 4,500 images

1.5 Train-Test Split

The pilot dataset was divided into training and testing sets using an 80:20 split. Stratified sampling was used to preserve the approximate class proportions in both sets. The random_state parameter was set to 42 so that the same split could be reproduced.

Training set: 3,600 images, shape (3600, 4096)

Testing set: 900 images, shape (900, 4096)

Class distribution in the training set:

Healthy: 1,273

Early Blight: 800

Late Blight: 1,527

Total: 3,600

Class distribution in the testing set:

Healthy: 318

Early Blight: 200

Late Blight: 382

Total: 900

1.6 Effect of Preprocessing and Data Splitting

Resizing standardised the image dimensions so that all images had the same number of features. Converting images to grayscale reduced the input from three colour channels to one channel, reducing dimensionality but also removing colour information that might help distinguish diseases. Flattening converted each two-dimensional image into a vector, making it suitable for conventional machine-learning algorithms but removing the explicit spatial arrangement of pixels. Pixel values were retained

in the range 0–255 at this stage; normalisation will be examined in a later experiment. The train-test split allows model performance to be evaluated on images not used for training.

1.7 Preliminary Verification Result

The prepared file was loaded successfully in VS Code. The feature matrix shape, label vector shape, pixel-value range, and class counts were verified programmatically. These checks confirmed that the pilot dataset was prepared in the expected format.

1.8 Limitation and Next Step

The initial pilot dataset contained 4,500 images from three tomato-leaf classes: Healthy, Early Blight and Late Blight. Since this dataset was insufficient for the required 5,000- and 10,000-image experiments, the dataset was expanded to include all 10 tomato classes available in the selected grayscale PlantVillage dataset. The expanded dataset contains 18,160 readable images and will be used for the final dataset-size experiments. The pilot results are reported separately and will not be treated as results from the expanded dataset.

Class	Total images	Training	Testing
Healthy	1,591	1,273	318
Early Blight	1,000	800	200
Late Blight	1,909	1,527	382
Total	4,500	3,600	900

1.9 Expanded Dataset Class Distribution

The expanded dataset contains 18,160 readable images belonging to 10 tomato-leaf classes. The class distribution was verified programmatically before conducting the dataset-size experiments.

Tomato-leaf class	Number of images
Bacterial Spot	2,127
Early Blight	1,000
Late Blight	1,909
Leaf Mold	952
Septoria Leaf Spot	1,771
Spider Mites	1,676

Target Spot	1,404
Yellow Leaf Curl Virus	5,357
Tomato Mosaic Virus	373
Healthy	1,591

Interpretation: The class distribution is imbalanced. Yellow Leaf Curl Virus has the highest number of images, while Tomato Mosaic Virus has the lowest. Therefore, accuracy alone may not fully represent how well a classifier performs on every class. Per-class precision, recall and F1-score will also be examined.

1.10 Dataset-Size Experiment Preparation

To investigate how training dataset size affects classification performance, the expanded dataset will be divided into training and testing sets using an 80:20 stratified split. A fixed random state of 42 will be used to make the split reproducible.

The training set will then be used to create four experimental training subsets.

Experiment	Training images
Dataset Size 1	1,000
Dataset Size 2	5,000
Dataset Size 3	10,000
Dataset Size 4 — Full available training set	14,528

The same held-out test set of 3,632 images will be retained for all four experiments. This ensures that differences in performance can be compared using the same test data while the amount of training data changes.

Interpretation: Comparing the four training sizes will help determine whether increasing the amount of training data improves classification performance and whether the improvement becomes smaller as more images are added. These are planned experiments; conclusions will be recorded only after the models have been trained and evaluated.

Dataset:

Expanded Dataset Selection: The initial three-class pilot dataset was expanded to include all 10 tomato-leaf classes. Details of the expanded dataset preparation and verification are provided in the following section.

Expanded Dataset Preparation and Verification

Each image was converted to grayscale and resized to 64 × 64 pixels. Therefore, each image contains 64 × 64 = 4,096 pixel features after flattening. The images are represented by a feature matrix X, and their corresponding class labels are stored in a label vector y.

The expanded dataset contains 18,160 readable images across 10 classes. The feature matrix has dimensions 18,160 × 4,096, and the label vector has dimensions 18,160. The prepared dataset was saved in compressed NumPy (.npz) format.

Interpretation: The expanded dataset provides enough images for the required training-size experiments. However, the class counts are unequal, so class imbalance must be considered when evaluating model performance.

Task 1: Data Representation and Vector Norms

1. Data representation

Each grayscale image is resized to 64×6464 \times 6464×64 pixels and flattened into a one-dimensional vector.

64×64=4096 features per image

The complete dataset is represented by a feature matrix X and a label vector y:

X∈R18160×4096

y∈Z18160

Here, each row of X represents one image, and each entry in y identifies its corresponding disease class.

2. Vector norms

L1 norm:

$$\|x\|_1 = \sum_{i=1}^{4096} |x_i|$$

L2 norm:

$$\|x\|_2 = \sqrt{\sum_{i=1}^{4096} x_i^2}$$

L-infinity norm

$$\|x\|_\infty = \max_{1 \le i \le 4096} |x_i|$$

Interpretation: The L1 norm measures the total absolute pixel intensity, the L2 norm measures the vector's Euclidean magnitude, and the L-infinity norm identifies the largest pixel value. These values describe the selected image numerically; they do not, by themselves, measure disease severity or classification accuracy.

Experimental results:

Norm	Calculated value
L1	402815.0000
L2	6747.9445
L-infinity	155.0000

Interpretation: The L1 norm represents the sum of the absolute pixel intensities, while the L2 norm represents the Euclidean magnitude of the image vector. The L-infinity norm indicates

that the maximum absolute pixel intensity in the selected image is 155. These measurements describe the numerical properties of the image rather than its disease severity.

Implementation: The norms were calculated using Python and NumPy. The image pixel values were converted to floating-point numbers before the calculations to avoid integer overflow during squaring.

Class	Images
Bacterial Spot	2,127
Early Blight	1,000
Late Blight	1,909
Leaf Mold	952
Septoria Leaf Spot	1,771
Spider Mites	1,676
Target Spot	1,404
Yellow Leaf Curl Virus	5,357
Tomato Mosaic Virus	373
Healthy	1,591
Total	18,160

Interpretation: The dataset is class-imbalanced because the number of images differs across disease categories. Yellow Leaf Curl Virus has the most images (5,357), while Tomato Mosaic Virus has the fewest (373). Therefore, accuracy alone may not fully represent performance across all classes. Precision, recall and F1-score will also be evaluated.

Task 2: Dimensionality Reduction Using PCA

2.1 Objective

The objective of this experiment is to investigate whether Principal Component Analysis (PCA) can reduce the number of features required to represent tomato-leaf images while preserving important information. Each image originally contains 4,096 pixel features. PCA is used to transform these features into a smaller number of principal components.

2.2 Methodology

The expanded dataset was divided into training and testing sets. Pixel values were normalized from the range 0–255 to 0–1. PCA was fitted using the training images only, with 100 principal components. The fitted transformation was then applied to a sample of 500 held-out test images to evaluate reconstruction error.

The experiment examined 10, 20, 50 and 100 principal components. Cumulative explained variance was used to measure how much of the variation in the training images was captured by the components. Reconstruction Mean Squared Error (MSE) was used to measure the difference between the original and reconstructed test images.

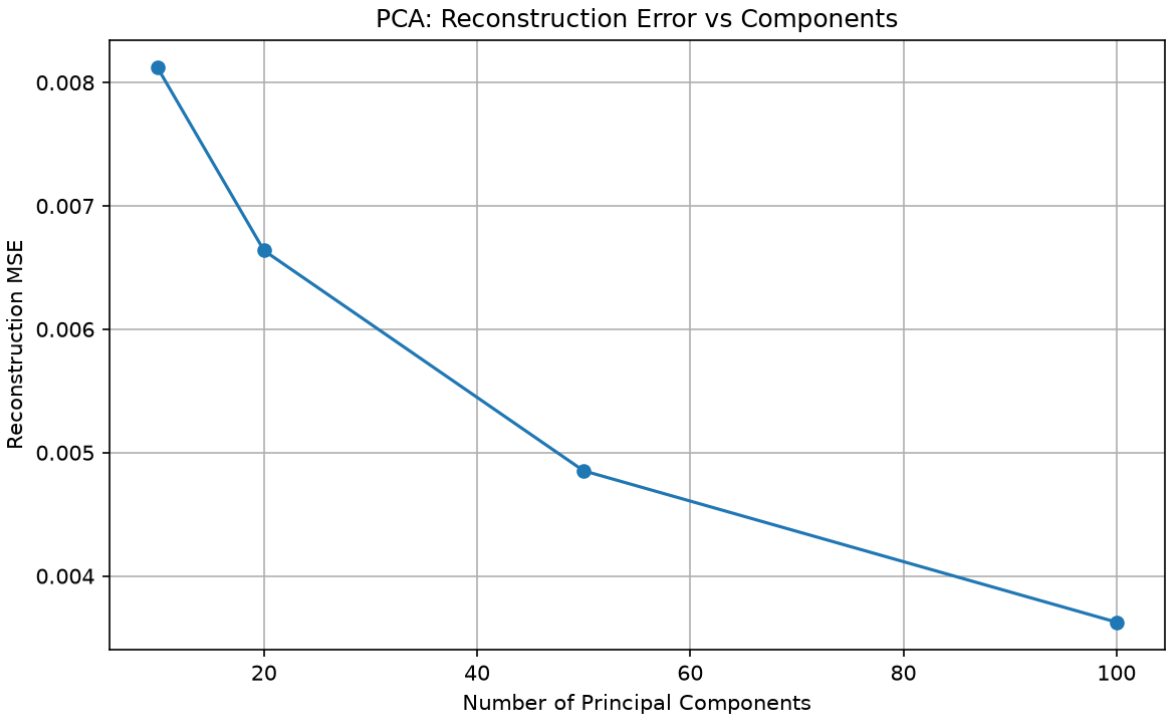
2.3 Experimental Results

Use this table with the actual output from your 08_pca_analysis.py program.

Principal components	Cumulative explained variance	Reconstruction MSE
10	60.6218%	0.008122
20	67.9343%	0.006638
50	76.8230%	0.004853
100	82.9080%	0.003626

Interpretation: As the number of principal components increases, cumulative explained variance increases and reconstruction error decreases. Using 100 components reduces the feature count from 4,096 to 100, a reduction of approximately 97.56%. However, explained variance alone does not establish whether classification accuracy improves.

2.4 Graphical Analysis



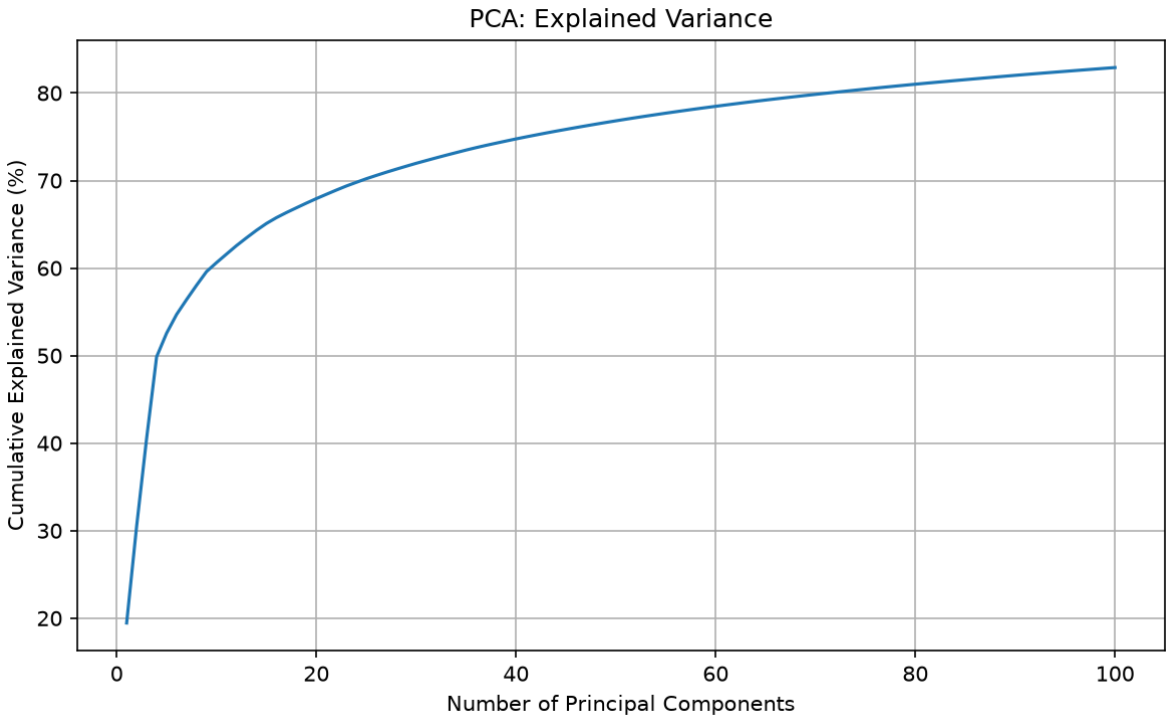


Figure 1: Cumulative explained variance against the number of principal components.

Figure 2: Reconstruction MSE against the number of principal components.

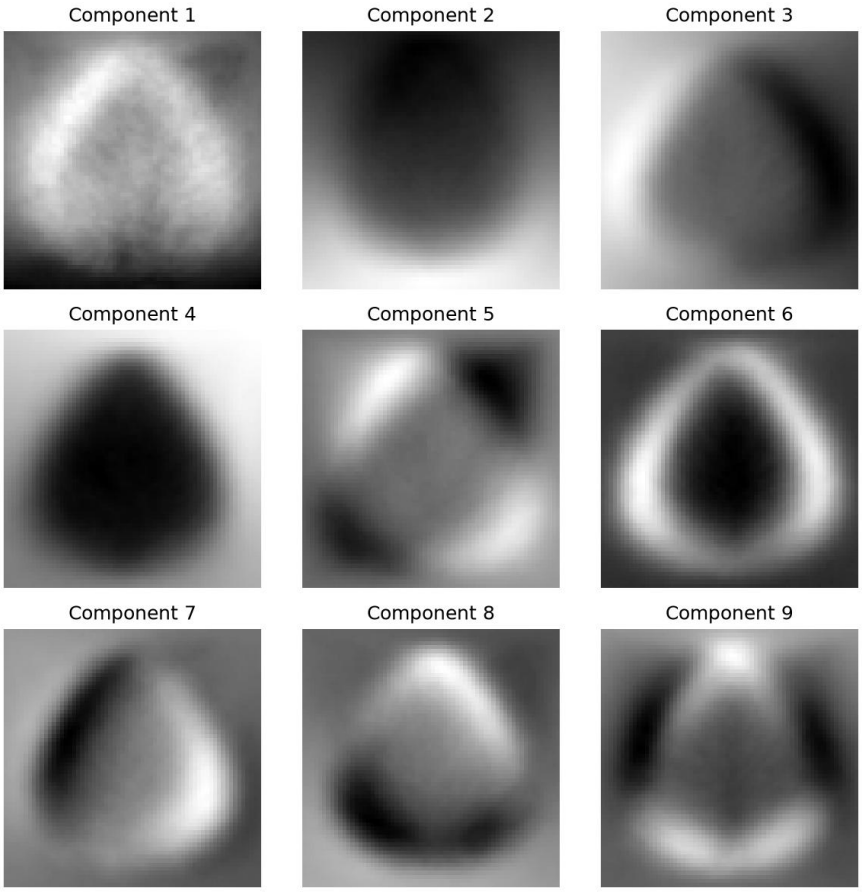


Figure 3: Visualization of the first nine principal components.

Interpretation: The explained-variance graph shows how the retained variance changes as more components are included. The reconstruction-error graph shows whether reconstructed images become closer to the original

images. The principal-component visualizations represent learned pixel patterns rather than actual disease photographs.

2.5 Mathematical Example of Eigenvalue Decomposition

Consider the matrix:

A= A=[2001]

Eigenvalues are obtained by solving:

det(A−λI)=0\det(A-\lambda I)=0det(A−λI)=0

Substituting the matrix gives:

|2−λ001−λ|=0\begin{vmatrix} 2-\lambda&0\\ 0&1-\lambda \end{vmatrix}=02−λ001−λ=0
(2−λ)(1−λ)=0(2-\lambda)(1-\lambda)=0(2−λ)(1−λ)=0

Therefore, the eigenvalues are:

λ1=2,λ2=1\lambda_1=2,\qquad \lambda_2=1λ1=2,λ2=1

For λ1=2\lambda_1=2λ1=2, an eigenvector is:

v1=[10]v_1=\begin{bmatrix}1\\0\end{bmatrix}v1=[10]

For λ2=1\lambda_2=1λ2=1, an eigenvector is:

v2=[01]v_2=\begin{bmatrix}0\\1\end{bmatrix}v2=[01]

Interpretation: In PCA, eigenvalues of a covariance matrix represent the variance associated with the corresponding principal directions. Larger eigenvalues indicate directions that capture more variance. This 2 × 2 example demonstrates the mathematical process; it is not a decomposition of the AgriVision dataset itself.

TASK 3 – Analyse Uncertainty Using Probability and Information Theory [K4]

3.1 Objective

The objective of this task is to examine uncertainty in AgriVision's class labels and model predictions using probability and information theory. Entropy, cross-entropy and Kullback–Leibler (KL) divergence are used to understand the effect of class imbalance, evaluate the quality of predicted probabilities and compare the confidence of different classifiers.

3.2 Class Distribution and Entropy

The expanded dataset contains 18,160 images belonging to 10 tomato-leaf classes. The number of images varies significantly between classes. Yellow Leaf Curl Virus has 5,357 images, whereas Tomato Mosaic Virus has only 373 images.

The probability of a class is calculated as:

$$ p_i=\frac{n_i}{N} $$

where n_i is the number of images in class i, and N is the total number of images.

The entropy of the class distribution is:

$$ H(Y)=-\sum_{i=1}^{10}p_i\log_2(p_i) $$

Entropy measures the uncertainty associated with the class label before considering the image features.

For ten equally represented classes, the maximum entropy is:

$$ H_{\max}=\log_2(10)\approx3.3219\text{ bits} $$

In AgriVision, the classes are not equally represented. Therefore, the class entropy is lower than the maximum possible value. The large proportion of Yellow Leaf Curl Virus images increases its contribution to the distribution, while the small number of Tomato Mosaic Virus images reduces that class's contribution.

However, lower entropy does not necessarily indicate a better dataset. Class imbalance may make a classifier favour majority classes and overlook minority diseases.

3.3 Cross-Entropy and KL Divergence

Cross-entropy measures the difference between the true class labels and the probabilities predicted by a model.

$$ H(P,Q)=-\sum_i p_i\log q_i $$

Here, P represents the true class distribution and Q represents the predicted probability distribution.

KL divergence measures how much the predicted distribution differs from the true distribution:

$$ D_{\mathrm{KL}}(P\parallel Q) =\sum_i p_i\log\left(\frac{p_i}{q_i}\right) $$

The relationship between these quantities is:

$$ H(P,Q)=H(P)+D_{\mathrm{KL}}(P\parallel Q) $$

This means that cross-entropy contains two components: the uncertainty inherent in the true distribution and the additional error caused by the predicted distribution.

For classification, the true label of each image is represented by a one-hot vector. If the correct class receives a predicted probability of 0.90, its cross-entropy loss is:

$$ L=-\ln(0.90)\approx0.1054 $$

If the correct class receives a probability of 0.20, the loss becomes:

$$ L=-\ln(0.20)\approx1.6094 $$

Therefore, assigning a low probability to the correct disease produces a larger penalty.

Illustrative comparison of two models

Consider a simplified two-class example, not an observed AgriVision experiment.

Let the true distribution be $P=(0.8,0.2)$. Suppose Model A predicts $Q_A=(0.7,0.3)$, while Model B predicts $Q_B=(0.5,0.5)$.

Measure	Model A	Model B
Cross-entropy (nats)	0.6390	0.6931
KL divergence (nats)	0.1382	0.1927

Model A has lower cross-entropy and lower KL divergence because its predicted probabilities are closer to the true distribution. For AgriVision, the same analysis can compare logistic regression, k-NN and tree-based models once their predicted probabilities have been obtained.

3.4 Why KL Divergence Is Asymmetric

KL divergence is asymmetric because:

$$ D_{\mathrm{KL}}(P\parallel Q) \ne D_{\mathrm{KL}}(Q\parallel P) $$

Using the same illustrative distributions:

$$ P=(0.8,0.2),\quad Q=(0.5,0.5) $$

The first direction is:

$$ D_{\mathrm{KL}}(P\parallel Q) =0.8\ln\frac{0.8}{0.5} +0.2\ln\frac{0.2}{0.5} $$

$$ D_{\mathrm{KL}}(P\parallel Q)\approx0.1927 $$

The reverse direction is:

$$ D_{\mathrm{KL}}(Q\parallel P) =0.5\ln\frac{0.5}{0.8} +0.5\ln\frac{0.5}{0.2} $$

$$ D_{\mathrm{KL}}(Q\parallel P)\approx0.2231 $$

The two values differ because the logarithmic ratios and their probability weights change when the distributions are reversed.

In the first direction, $D_{\mathrm{KL}}(P\parallel Q)$ measures the information lost when the predicted distribution Q is used to represent the true distribution P. In the reverse direction, $D_{\mathrm{KL}}(Q\parallel P)$ evaluates the discrepancy with the distributions interchanged.

3.5 Why Minimising Cross-Entropy Also Minimises KL Divergence

For a fixed true distribution P,

$$ H(P,Q)=H(P)+D_{\mathrm{KL}}(P\parallel Q) $$

Since $H(P)$ is constant with respect to the model's parameters, minimising cross-entropy is equivalent to minimising $D_{\mathrm{KL}}(P\parallel Q)$.

This explains why cross-entropy is widely used as the loss function for classification models. Minimising it encourages the model to assign higher probabilities to the correct disease classes.

3.6 Interpretation

The class imbalance identified in the original dataset may cause a classifier to achieve reasonable overall accuracy while performing poorly on minority classes. Entropy describes

the uncertainty of the class distribution, while cross-entropy and KL divergence measure the mismatch between true and predicted probabilities.

A meaningful comparison must therefore examine not only accuracy but also per-class precision, recall, F1-score and predicted probability quality. The model with the lowest measured cross-entropy is better calibrated to the labels under that metric, although it is not necessarily the model with the highest classification accuracy.

TASK 4 – Diagnose Numerical Computation Issues [K4]

4.1 Objective

The objective of this task is to investigate numerical failures that can occur when calculating class probabilities and gradients. The analysis focuses on softmax overflow, underflow, numerical stabilisation and the effect of the feature covariance matrix's condition number on optimisation.

4.2 Softmax Overflow and Underflow

Softmax converts a vector of logits into class probabilities:

$$p_i=\frac{e^{z_i}}{\sum_j e^{z_j}}$$

Consider the logits:

$$z=[1000,999,995]$$

Direct computation requires evaluating e^{1000}, e^{999} and e^{995}. These values exceed the representable range of standard floating-point numbers, resulting in overflow. The denominator may become infinity, and the calculated probabilities may become undefined.

Underflow can occur when logits are extremely negative. Their exponentials may become zero in floating-point arithmetic. If every exponential becomes zero, the denominator is zero and the probability calculation fails.

4.3 Stable Softmax Using Maximum Subtraction

The numerically stable form of softmax is:

$$p_i=\frac{e^{z_i-m}}{\sum_j e^{z_j-m}}$$

where:

$$ m=\max_j z_j $$

For the example:

$$ z-m=[0,-1,-5] $$

The probabilities are then calculated from:

$$ e^0=1,\quad e^{-1}\approx0.3679,\quad e^{-5}\approx0.0067 $$

Therefore:

$$ \sum_j e^{z_j-m}\approx1.3746 $$

The resulting probabilities are approximately:

$$ P=[0.7275,0.2676,0.0049] $$

The result is mathematically equivalent to the original softmax because the common factor e^{-m} cancels between the numerator and denominator. Subtracting the maximum prevents the largest exponent from overflowing and improves numerical stability.

4.4 The Log-Sum-Exp Trick

The logarithm of the sum of exponentials can be calculated as:

$$ \log\sum_i e^{z_i} =m+\log\sum_i e^{z_i-m} $$

This is known as the log-sum-exp trick. It avoids directly calculating very large exponentials and is particularly useful when computing softmax cross-entropy loss.

For classification, stable log probabilities should be calculated directly rather than taking the logarithm of probabilities that may have underflowed to zero.

4.5 Condition Number of the Feature Covariance Matrix

The condition number describes the ratio between the largest and smallest positive eigenvalues of a matrix:

$$ \kappa(\Sigma)=\frac{\lambda_{\max}}{\lambda_{\min}} $$

A large condition number indicates that some feature directions vary much more strongly than others. In optimisation, this can produce elongated loss contours, causing gradient descent to move rapidly along some directions and slowly along others.

AgriVision contains 4,096 pixel features per image. Pixel features may be correlated, and some directions may have very small variance. These properties can make the covariance matrix ill-conditioned.

PCA identifies directions of high variance and can reduce the number of features. However, PCA does not automatically guarantee a well-conditioned optimisation problem. The condition number must be measured on the relevant transformed features, and zero-variance directions require special treatment.

4.6 Interpretation

Naive softmax may fail because floating-point arithmetic cannot represent extremely large or extremely small exponential values. Maximum subtraction and log-sum-exp address this issue without changing the mathematical probabilities.

An ill-conditioned covariance matrix may slow or destabilise gradient-based learning. Consequently, numerical stability, feature scaling and appropriate learning-rate selection are important for reliable AgriVision model training. The exact condition number and its effect on training speed must be determined experimentally.

TASK 5 – Differentiate Baseline Learning Strategies [K4]

5.1 Objective

The objective is to compare three distinct machine-learning algorithms on AgriVision's raw image features and PCA-compressed features. The algorithms differ in their hypothesis spaces, assumptions and methods of learning.

The selected algorithms are:

1. Logistic Regression — a linear parametric classifier.
2. K-Nearest Neighbours (k-NN) — a non-parametric, instance-based classifier.
3. Extra Trees — a tree-based ensemble classifier.

Each model should be trained on the same training images and evaluated using the same held-out test set to ensure a fair comparison.

5.2 Comparison of Learning Strategies

Characteristic	Logistic Regression	k-NN	Extra Trees
Learning strategy	Linear parametric	Distance-based	Tree ensemble
Hypothesis space	Linear decision boundaries	Local class assignments	Nonlinear partitions
Training behaviour	Learns model coefficients	Stores training examples	Learns many decision trees
Main assumption	Classes can be separated approximately by linear boundaries in the chosen feature space	Nearby examples tend to have similar labels	Useful class patterns can be captured by feature-based splits
Main strength	Efficient prediction and interpretable coefficients	Flexible local decisions	Captures nonlinear relationships
Main limitation	May miss complex image patterns	Sensitive to distance, scaling and high dimensionality	Can require substantial memory and computation

5.3 Effect of Raw 4,096-Dimensional Features

Each image is represented by 4,096 grayscale pixel values. Although this representation preserves pixel intensity information, flattening removes the explicit two-dimensional arrangement of pixels.

Logistic Regression learns a weighted combination of the pixel features. It is relatively efficient, but its linear decision boundaries may be insufficient to distinguish complex disease patterns.

k-NN compares a test image with stored training images using a distance measure. In high-dimensional spaces, distances may become less discriminative, and irrelevant pixel variations can affect the nearest neighbours. Its inference cost also increases with the number of stored training examples.

Extra Trees learns nonlinear decision boundaries using multiple randomised trees. It can model interactions between pixels, but trees do not inherently understand spatial structure. Deep trees may overfit, while large ensembles can consume substantial memory.

5.4 Effect of PCA Compression

The original 4,096 features can be reduced to 100 principal components, corresponding to a feature reduction of approximately:

\[\frac{4096-100}{4096}\times100 \approx97.56\% \]

PCA compression affects the three algorithms differently.

- Logistic Regression: Fewer features reduce the number of coefficients and may reduce training cost. However, removing low-variance directions may also remove useful class information.
- k-NN: Distance calculations become cheaper because each comparison uses fewer features. However, distances between projected images may not preserve every distinction needed for disease classification.
- Extra Trees: Fewer features can reduce the number of candidate splits and training overhead. Nevertheless, the principal components combine information from multiple original pixels, which may make some splits less directly interpretable.

5.5 Experimental Comparison

The following table should be completed using the actual measured results.

Algorithm	Feature representation	Test accuracy	Macro F1-score	Training time
Logistic Regression	Raw 4,096 features	To be measured	To be measured	To be measured
Logistic Regression	PCA, 100 components	To be measured	To be measured	To be measured
k-NN	Raw 4,096 features	To be measured	To be measured	To be measured

k-NN	PCA, 100 components	To be measured	To be measured	To be measured
Extra Trees	Raw 4,096 features	To be measured	To be measured	To be measured
Extra Trees	PCA, 100 components	To be measured	To be measured	To be measured

5.6 Interpretation

The theoretical comparison suggests that PCA may particularly benefit distance-based methods by reducing the cost of comparing high-dimensional images. Logistic Regression may benefit from reduced computation, while Extra Trees may gain from a smaller feature space but could lose information needed for useful decision splits.

These are hypotheses, not established rankings. The best approach must be determined from test accuracy, macro F1-score, inference time and memory usage. Macro F1-score is especially important because the original dataset contains substantially different class counts.

TASK 6 – Analyse Capacity, Overfitting and Underfitting [K4]

6.1 Objective

This task investigates how model capacity affects training error, validation error and generalisation. Model capacity refers to the complexity of the patterns a model can represent.

Three capacity controls are considered:

- Logistic Regression: regularisation strength.
- k-NN: number of neighbours, \(k\).
- Extra Trees: maximum tree depth.

6.2 Effect of Capacity on Training and Validation Error

Logistic Regression

Regularisation controls the size of the model coefficients. Strong regularisation restricts the model's flexibility, whereas weak regularisation allows it to fit the training data more closely.

A common regularised objective is:

$$ J(w)= -\frac{1}{N}\sum_{i=1}^{N}\log p(y_i\mid x_i;w) +\lambda\|w\|_2^2 $$

Here, λ controls the regularisation strength. Larger values penalise large coefficients more strongly.

If regularisation is excessive, the model may underfit. If it is too weak, the model may become more sensitive to noise and produce a larger generalisation gap.

k-Nearest Neighbours

The parameter k determines how many training examples contribute to a prediction.

- Small k: highly local decisions, low training error may be possible, but sensitivity to noise is high.
- Large k: smoother decisions and greater averaging, but small or unusual classes may be overwhelmed by neighbouring majority classes.

A very small k can overfit, while an excessively large k can underfit.

Extra Trees

Maximum tree depth controls how complex individual decision trees can become.

- Shallow trees may not capture important class differences.
- Deeper trees can capture more complex relationships.
- Very deep trees may fit noise and peculiarities of the training images.

An ensemble can reduce some variance compared with an individual tree, but increasing tree depth does not automatically improve generalisation.

6.3 Underfitting and Overfitting

Underfitting — low capacity

Both training and validation performance are poor because the model cannot represent enough of the relationship between image features and disease labels.

Good generalisation — suitable capacity

The model captures useful patterns while maintaining similar training and validation performance.

Overfitting — excessive capacity

Training performance improves, but validation performance stagnates or deteriorates because the model learns noise or details that do not generalise.

6.4 Generalisation Gap

The generalisation gap can be expressed as:

$$ G=E_{\mathrm{validation}}-E_{\mathrm{training}} $$

where E represents classification error.

A small gap with low training error suggests good generalisation. A large positive gap indicates that the model performs substantially worse on unseen validation images than on its training images.

However, a small gap does not automatically mean that a model is good: both errors can be high when the model underfits.

6.5 Capacity–Error Experiment

The following experiments should be performed using the training set and a separate validation set.

Model	Capacity settings to compare	Main question
Logistic Regression	Different regularisation strengths	Does weaker regularisation improve fit or increase the generalisation gap?
k-NN	$k=1,3,5,11,21$	Does increasing k reduce variance at the cost of higher bias?
Extra Trees	Maximum depths of 5, 10, 20 and unrestricted	Does greater depth improve validation performance or mainly reduce training error?

Plot training error and validation error against the relevant capacity setting. Mark the underfitting region, the best validation region and the overfitting region only after observing the measured curves.

6.6 Why the Test Set Must Not Select Hyperparameters

Repeatedly selecting hyperparameters based on test performance introduces information from the test set into the model-development process. The reported test error then becomes an optimistic estimate of performance on genuinely unseen data.

A separate validation set or k-fold cross-validation should be used for model selection. The test set should be reserved for the final evaluation after the modelling decisions have been completed.

6.7 Interpretation

Model capacity creates a trade-off between fitting the training data and generalising to unseen images. In AgriVision, excessive capacity may allow a model to memorise image-specific noise, while insufficient capacity may prevent it from distinguishing visually similar diseases.

The optimal capacity is the one that achieves the best validation performance without an unnecessarily large generalisation gap. The precise underfitting and overfitting regions must be identified from the experimental curves.

TASK 7 – Decompose Error into Bias and Variance [K4]

7.1 Objective

The objective is to investigate how model complexity contributes to prediction error and how prior knowledge affects parameter estimation. A bootstrap experiment is used to compare low- and high-capacity models, followed by a comparison of maximum likelihood estimation (MLE) and maximum a posteriori estimation (MAP).

7.2 Bias–Variance Decomposition

For a regression prediction with squared-error loss, expected prediction error can be expressed as:

\[E[(Y-\hat f(X))^2] = \operatorname{Bias}[\hat f(X)]^2 +\operatorname{Var}[\hat f(X)] +\sigma^2 \]

The terms are:

- Bias²: error caused by systematic limitations of the model.
- Variance: error caused by changes in the training sample.
- Noise: irreducible variation in the target.

A low-capacity model may have high bias because it cannot capture complex patterns. A high-capacity model may have lower bias but higher variance because its predictions change more strongly when the training data changes.

For multiclass classification, this exact squared-error decomposition does not directly apply to 0–1 classification error. A practical alternative is to use the multiclass Brier score and explicitly identify it as a probability-based squared-error measure.

7.3 Bootstrap Experiment

Bootstrap sampling creates new training samples by sampling the original training data with replacement. At least ten bootstrap replicas should be used for each model.

The experimental procedure is:

1. Select a fixed evaluation set.
2. Create ten or more bootstrap training samples.
3. Train a low-capacity model on each sample.
4. Train a high-capacity model on the same bootstrap samples.
5. Save the predicted class-probability vector for each evaluation image.
6. Calculate average predictions and the variability of predictions across replicas.
7. Compare the resulting bias-related and variance-related measurements.

For probability prediction $\hat p_b(x)$ from bootstrap model b, the average prediction is:

$$\bar p(x)=\frac{1}{B}\sum_{b=1}^{B}\hat p_b(x)$$

where $B\geq10$.

The variation across bootstrap models indicates how sensitive the prediction is to the training sample. The bias-related component measures the discrepancy between the average prediction and the true class target, under the selected loss definition.

7.4 Experimental Results Table

Model capacity	Number of replicas	Bias-related error	Variance-related error	Total measured error

Low capacity	At least 10	To be measured	To be measured	To be measured
High capacity	At least 10	To be measured	To be measured	To be measured

For a Brier-score-based analysis, the theoretical decomposition should be stated consistently with the target representation and noise assumptions. Any difference between the measured error and the estimated components may arise from finite bootstrap samples, approximation choices and differences between the evaluation population and the observed dataset.

7.5 MLE and MAP Estimation of a Class Proportion

Suppose a binary event is observed s times in n trials, and its probability is θ.

The likelihood is proportional to:

$$ L(\theta)\propto\theta^s(1-\theta)^{n-s} $$

The maximum likelihood estimate is:

$$ \hat\theta_{\mathrm{MLE}}=\frac{s}{n} $$

Now assume a Beta prior:

$$ \theta\sim\operatorname{Beta}(\alpha,\beta) $$

The posterior distribution is:

$$ \theta\mid D\sim \operatorname{Beta}(\alpha+s,\beta+n-s) $$

For $\alpha>1$ and $\beta>1$, the MAP estimate is:

$$ \hat\theta_{\mathrm{MAP}} =\frac{s+\alpha-1}{n+\alpha+\beta-2} $$

Numerical example

Suppose 8 out of 10 examined images belong to a particular binary category, and the prior is $\operatorname{Beta}(2,2)$.

The MLE is:

$$ \hat\theta_{\mathrm{MLE}}=\frac{8}{10}=0.8 $$

The MAP estimate is:

$$ \hat\theta_{\mathrm{MAP}} =\frac{8+2-1}{10+2+2-2} =\frac{9}{12}=0.75 $$

The prior pulls the MAP estimate towards 0.5, reducing the influence of a small sample.

7.6 Effect of Sample Size

For the same observed proportion of 0.8 and the same prior:

Successes / observations	MLE	MAP
8 / 10	0.8000	0.7500
80 / 100	0.8000	0.7941
800 / 1,000	0.8000	0.7994

As the sample size increases, the prior contributes a smaller proportion of the total evidence, and the MLE and MAP estimates become closer.

7.7 Interpretation

Bootstrap analysis helps distinguish models that are systematically limited from models that are sensitive to changes in the training data. The expected pattern is that simpler models may have greater bias, while more flexible models may show greater prediction variance.

The MLE depends entirely on observed data, whereas MAP combines observed evidence with a prior distribution. For AgriVision, prior information may be useful when estimating proportions from small minority classes, but the prior should be chosen carefully and must not replace empirical validation.

The exact bias–variance measurements for the selected image classifiers remain to be established experimentally.

TASK 8 – Analyse the Behaviour of Gradient-Based Optimisation [K4]

8.1 Objective

This task investigates how optimisation strategies and learning rates affect the training of softmax regression on AgriVision's image features. The gradient is derived mathematically, and batch gradient descent, stochastic gradient descent and mini-batch gradient descent are compared.

8.2 Softmax Regression Model

Let the input feature vector be:

$$ x_i\in\mathbb{R}^{d} $$

For the raw AgriVision images, $d=4096$. Let $K=10$ represent the number of disease classes.

The model computes the logits:

$$ z_i=W^Tx_i+b $$

where W contains the model weights and b contains the class biases.

The predicted probability for class k is:

$$ p_{ik}=\frac{e^{z_{ik}}}{\sum_{j=1}^{K}e^{z_{ij}}} $$

For a one-hot target y_i, the cross-entropy loss for a single image is:

$$ L_i=-\sum_{k=1}^{K}y_{ik}\log p_{ik} $$

The average loss over N training images is:

$$ J(W,b)=-\frac{1}{N} \sum_{i=1}^{N}\sum_{k=1}^{K} y_{ik}\log p_{ik} $$

An optional regularisation term may be added to control model complexity.

8.3 Derivation of the Gradient

The derivative of the softmax probability with respect to a logit is:

$$ \frac{\partial p_k}{\partial z_j} =p_k(\delta_{kj}-p_j) $$

where δ_{kj} is 1 when $k=j$, and 0 otherwise.

Combining softmax with cross-entropy simplifies the derivative to:

$$ \frac{\partial L_i}{\partial z_{ik}} =p_{ik}-y_{ik} $$

Since $z_{ik}=w_k^Tx_i+b_k$, the chain rule gives:

$$ \frac{\partial L_i}{\partial w_k} =(p_{ik}-y_{ik})x_i $$

Therefore, the gradient of the average loss with respect to the weights is:

$$ \boxed{ \nabla_W J=\frac{1}{N}X^T(P-Y) } $$

The gradient with respect to the bias vector is:

$$ \boxed{ \nabla_b J=\frac{1}{N}\sum_{i=1}^{N}(p_i-y_i) } $$

These gradients show that each update depends on the difference between predicted probabilities and true labels. Errors in probability estimates produce updates that change the weights in directions intended to reduce the loss.

For L2 regularisation with coefficient λ, the weight gradient gains an additional term proportional to W, depending on the chosen regularisation convention.

8.4 Optimisation Strategies

Batch Gradient Descent

Batch gradient descent calculates the gradient using the entire training set before making an update.

$$ W_{t+1}=W_t-\eta\nabla_WJ $$

It produces a relatively stable gradient direction but requires processing all training examples for each update. With 14,528 training images and 4,096 features per image, each full-batch update may be computationally expensive.

Stochastic Gradient Descent

Stochastic gradient descent updates the parameters using one training example at a time.

$$ W_{t+1}=W_t-\eta\nabla_WL_i $$

It performs inexpensive individual updates but introduces substantial gradient noise. This can help the model explore the loss surface, although the loss may fluctuate and convergence may be less smooth.

Mini-Batch Gradient Descent

Mini-batch SGD calculates each update using a small group of training examples.

$$ W_{t+1}=W_t-\eta \frac{1}{B}\sum_{i\in\mathcal B_t} \nabla_WL_i $$

where B is the mini-batch size.

Mini-batch SGD balances computational efficiency and gradient stability. It is often a practical choice for image classification.

8.5 Learning-Rate Analysis

The learning rate η determines the size of each parameter update.

Learning rate	Expected behaviour	Potential problem
0.001	Small, cautious updates	Slow convergence
0.01	Larger updates that may improve training speed	May require monitoring for oscillation
0.1	Aggressive updates	May overshoot the minimum or diverge

These are candidate learning rates for an experiment, not measured outcomes. Their behaviour depends on feature scaling, the loss surface, regularisation and the optimiser.

A very small learning rate can require many iterations to make meaningful progress. A very large learning rate can cause the loss to fluctuate or increase instead of converging.

8.6 Connection to Numerical Conditioning

Task 4 examined how a large covariance-matrix condition number can create directions with very different curvature. When this occurs, a learning rate suitable for a steep direction may be too small for a shallow direction, while a rate suitable for the shallow direction may cause instability along the steep direction.

PCA reduces the number of input dimensions, which may reduce computation and remove some low-variance directions. However, the effect on conditioning must be measured rather than assumed. PCA's explained variance alone does not prove that optimisation will become faster.

8.7 Experimental Evidence

The following measurements should be recorded for each optimisation strategy and learning rate:

Optimiser	Learning rate	Final training loss	Validation loss	Convergence time
Batch GD	0.001	To be measured	To be measured	To be measured
Batch GD	0.01	To be measured	To be measured	To be measured

Batch GD	0.1	To be measured	To be measured	To be measured
SGD	0.001, 0.01, 0.1	To be measured	To be measured	To be measured
Mini-batch SGD	0.001, 0.01, 0.1	To be measured	To be measured	To be measured

Plot training and validation loss against epochs or elapsed time. The x-axis should be consistent within each comparison, and the legend should identify the optimiser and learning rate.

8.8 Interpretation

Batch gradient descent provides full-dataset gradients but can be expensive per update. SGD makes rapid individual updates but has noisy convergence, while mini-batch SGD balances these behaviours.

The most suitable strategy for AgriVision is the one that reaches a low validation loss efficiently without numerical instability. The final conclusion should be based on measured loss curves, runtime and conditioning rather than assuming that a particular optimiser or learning rate is always best.

TASK 9 – Organise and Examine Experimental Performance [K4]

9.1 Objective

The objective is to organise the performance of the three baseline classifiers across increasing training dataset sizes. Accuracy, precision, recall, F1-score, confusion matrices, training time, inference time, parameter count and memory footprint are considered.

The original report specifies four training sizes: 1,000, 5,000, 10,000 and 14,528 images. All four experiments should use the same held-out test set of 3,632 images.

9.2 Evaluation Metrics

Accuracy

Accuracy is the proportion of correctly classified test images:

$$\mathrm{Accuracy}=\frac{TP+TN}{TP+TN+FP+FN}$$

For multiclass classification, it is the number of correct predictions divided by the total number of test images.

Precision

$$\mathrm{Precision}=\frac{TP}{TP+FP}$$

Precision measures how many images predicted as a particular disease actually belong to that disease.

Recall

$$\mathrm{Recall}=\frac{TP}{TP+FN}$$

Recall measures how many actual images of a disease are correctly identified.

F1-score

$$F1=2\frac{\mathrm{Precision}\times\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}}$$

Macro-averaged precision, recall and F1-score are calculated by evaluating each class separately and taking the unweighted mean across classes. This prevents the largest class from dominating the summary metric.

9.3 Dataset-Size Performance Table

The following table should be populated with measured values.

Training images	Model	Accuracy	Macro precision	Macro recall	Macro F1
1,000	Logistic Regression	To be measured	To be measured	To be measured	To be measured
1,000	k-NN	To be measured	To be measured	To be measured	To be measured
1,000	Extra Trees	To be measured	To be measured	To be measured	To be measured
5,000	All three models	To be measured	To be measured	To be measured	To be measured
10,000	All three models	To be measured	To be measured	To be measured	To be measured

14,528	All three models	To be measured	To be measured	To be measured	To be measured	To be measured	To be measured

The full experiment should contain one result row per model, feature representation and dataset size. Raw and PCA-compressed representations should be reported separately.

9.4 Confusion Matrix Analysis

A confusion matrix compares the true labels with the predicted labels. Each row represents an actual class, and each column represents a predicted class.

Correct classifications appear on the diagonal. Off-diagonal values identify disease classes that the model confuses with one another.

For example, if Early Blight is repeatedly predicted as Late Blight, the confusion matrix will reveal this pattern. However, the actual AgriVision confusion matrix must be examined before identifying any particular pair as frequently confused.

Possible reasons for confusion include similar visual symptoms, overlapping lesion patterns, changes in illumination, background differences and information lost through grayscale conversion or PCA compression.

9.5 Computational Cost and Memory

The following measurements should be recorded for every model and feature representation.

Measure	**What it indicates**
Training time	Time required to fit the model
Inference time per image	Time needed to predict one image
Parameter count	Number of learned coefficients or model parameters, where applicable
Model size	Storage required by the saved model
Peak memory	Maximum memory used during training or inference

Parameter count and model size are not identical. For example, a tree ensemble stores split information and tree structures rather than a fixed dense matrix of coefficients. For k-NN, the stored training examples may dominate memory usage.

Inference timing should be performed on the same machine under comparable conditions, with warm-up and repeated measurements where appropriate.

9.6 Expected Dataset-Size Patterns

Increasing the training set may improve performance because the model receives more examples of the disease classes. The benefit may be especially important for classes that have few images.

However, improvement is not guaranteed at every size. Larger datasets require more training time and, for some methods, more memory. If the additional images contain substantial redundancy or noise, their marginal benefit may be small.

PCA compression may reduce computational cost, but a decrease in feature count can also remove information relevant to classification.

9.7 Interpretation

The final comparison should identify which model provides the best balance between predictive performance, minority-class recognition, training cost, inference latency and memory.

A model should not be selected solely because it achieves the highest accuracy. In a disease-screening application, low recall for a particular disease may be an important limitation even when overall accuracy is high.

TASK 10 – Compare Theoretical Expectations with Experimental Behaviour [K4]

10.1 Objective

This task evaluates whether the measured AgriVision results agree with the theoretical expectations established in Tasks 1–9. It connects PCA, numerical conditioning, optimisation, capacity and bias–variance analysis.

10.2 PCA Explained Variance Versus Classification Accuracy

The original report provides these PCA results:

Components	Cumulative explained variance	Reconstruction MSE
10	60.6218%	0.008122
20	67.9343%	0.006638
50	76.8230%	0.004853
100	82.9080%	0.003626

These values show that more components preserve more of the training-image variance and reduce reconstruction error in the reported experiment.

However, explained variance does not directly measure disease-discriminative information. A component capturing large lighting or background variation may explain substantial variance without improving disease classification. Conversely, a low-variance direction may contain useful information for distinguishing two visually similar diseases.

Therefore, the classification accuracy and macro F1-score of each PCA representation must be compared with the raw-feature baseline.

10.3 Optimisation Speed Versus Conditioning

Task 4 predicts that poorly conditioned features may slow gradient-based optimisation or make it more sensitive to the learning rate.

If the measured covariance condition number decreases after preprocessing and convergence becomes faster, the results support this expectation. If the condition number decreases but convergence does not improve, possible explanations include the overhead of the transformation, the optimiser configuration, the loss function or the removal of useful features.

The relationship must be demonstrated with measured condition numbers and comparable optimisation experiments.

10.4 Bias–Variance Results Versus Capacity Curves

Task 6 examines how capacity changes training and validation error, while Task 7 estimates bias-related error and prediction variance across bootstrap samples.

A simple model that performs poorly on both training and validation data is consistent with underfitting and high bias. A complex model with low training error, substantially worse

validation performance and variable bootstrap predictions is consistent with overfitting and high variance.

Any disagreement may arise because bootstrap variability, classification error and probability-based losses measure different aspects of model performance. The same loss definition and evaluation procedure must be used when making quantitative comparisons.

10.5 Sources of Disagreement

Differences between theory and measured results may occur because of:

- Class imbalance and unequal sample sizes.
- Loss of spatial information during image flattening.
- Information removed by PCA.
- Differences in model capacity and regularisation.
- Numerical precision and optimisation settings.
- Randomness in training or model construction.
- Variation between the sampled dataset and the wider population of crop images.

10.6 Interpretation

Mathematical analysis provides expectations about model behaviour, but experimental evidence determines whether those expectations hold for the chosen dataset and implementation.

The AgriVision design should therefore be evaluated by comparing theoretical predictions with measured accuracy, macro F1-score, training and inference time, conditioning and generalisation gap. Disagreements should be investigated rather than hidden, as they may reveal limitations in the representation or experimental setup.

TASK 11 – Analyse Whether Deep Learning Is Justified [K4]

11.1 Objective

The objective is to determine whether the limitations of classical machine-learning algorithms justify the use of a deep-learning model for AgriVision. The decision must be based on

evidence from the baseline experiments, PCA analysis, computational requirements and the characteristics of crop-disease images.

11.2 Limitations of Classical Machine Learning

The original dataset represents every image as a flattened vector of 4,096 grayscale pixel values. This representation is convenient for conventional algorithms, but it does not explicitly preserve the two-dimensional spatial relationships between pixels.

Logistic Regression learns linear decision boundaries in the chosen feature space. It may struggle to distinguish diseases whose visual patterns require complex nonlinear relationships.

k-NN depends on the similarity between stored images and a query image. Its performance may deteriorate when high-dimensional distances become less informative, and its inference cost increases as the number of stored training examples grows.

Extra Trees can learn nonlinear decision boundaries, but the individual trees still rely on feature-based splits rather than automatically learning a hierarchy of spatial image features. Large ensembles may also create memory and inference-time challenges.

These limitations provide reasons to investigate deep learning, but they do not establish that a neural network will necessarily perform better.

11.3 Curse of Dimensionality

The curse of dimensionality refers to difficulties that arise as the number of features increases. In high-dimensional spaces, data may become sparse relative to the available volume, and distance-based methods may find it difficult to identify meaningful neighbourhoods.

AgriVision has 4,096 input features per image. This is a substantial feature space compared with the number of training images in the smaller experiments.

PCA reduces the dimensionality to as few as 100 components, which can lower computational costs. Nevertheless, the retained components must be evaluated according to classification performance, not merely explained variance.

11.4 Local Constancy and Smoothness

k-NN assumes that sufficiently nearby images tend to have similar class labels. This is a local-constancy assumption.

However, two images of the same disease can differ because of illumination, camera angle, leaf orientation, background or disease progression. Conversely, different diseases may produce visually similar symptoms.

The distance between flattened pixel vectors may not always represent these meaningful similarities. A representation that is invariant to irrelevant changes and sensitive to disease-specific patterns may therefore improve classification.

11.5 Manifold Structure and Hierarchical Representation

Crop-disease images contain complex visual patterns, including leaf shapes, lesion boundaries, spots, textures and colour or intensity changes. Although the raw pixel space has 4,096 dimensions, meaningful leaf images may occupy a much more structured region of that space.

A convolutional neural network can learn hierarchical representations, progressing from local edges and textures to more complex disease-related patterns. Weight sharing and local receptive fields also provide useful assumptions for image data.

These properties make CNNs a reasonable candidate for AgriVision, particularly when spatial structure is important. However, the existence and usefulness of a low-dimensional manifold must not be treated as experimentally proven solely from the dataset's dimensionality.

11.6 Evidence Needed to Justify Deep Learning

A deep-learning model should be compared with the best tuned classical baseline using:

- Accuracy and macro F1-score on the same held-out test set.
- Per-class recall, especially for underrepresented diseases.
- Training time and inference latency.
- Model size and peak memory usage.
- Robustness to changes in illumination, image backgrounds and label noise.

If the best classical model still exhibits substantial confusion between diseases, poor minority-class recall or inadequate performance under realistic image variations, a learned hierarchical representation may be justified.

If a PCA-compressed classical model achieves comparable predictive performance with substantially lower latency and memory usage, deep learning may not be necessary for the current requirements.

11.7 Trade-Offs

Factor	Classical machine learning	Deep learning
Data requirement	Can be effective with moderate datasets, depending on representation	Often benefits from larger and more diverse datasets
Feature extraction	Frequently relies on raw pixels or manually designed representations	Learns hierarchical features from training data
Computation	Can be relatively efficient, depending on the model	Training can be expensive; inference varies by architecture
Interpretability	Some models offer relatively direct explanations	Learned representations are generally harder to interpret
Spatial image patterns	Must be inferred indirectly from the selected features	CNNs are specifically designed to exploit local spatial structure

11.8 Conclusion

Deep learning is a justified candidate for further evaluation because AgriVision is an image-classification problem with high-dimensional inputs and potentially complex visual patterns. However, the existing report does not yet contain the completed baseline results needed to prove that classical machine learning is inadequate.

The final decision should therefore be based on an empirical comparison between the best classical model and a suitably designed CNN. Deep learning should be selected only if the measured improvement in classification quality or robustness justifies its computational and interpretability costs.

TASK 12 – Analyse the Impact of Changed Requirements [K4]

12.1 Changed Requirement

The revised AgriVision system must classify 38 crop–disease classes, handle approximately 20% noisy labels, run on a low-cost smartphone within 50 milliseconds per image and occupy no more than 5 MB of model memory.

These requirements introduce changes to the original problem's class space, training-label reliability, inference latency and memory budget.

The original dataset contains 10 tomato-leaf classes and 18,160 readable images. It does not establish performance on 38 crop–disease classes. Consequently, the revised design must be evaluated as a new deployment scenario rather than assumed to work under the original conditions.

A. Impact Analysis [2 Marks]

A.1 Effect of Increasing the Number of Classes

For K equally probable classes, the maximum class entropy is:

$$ H_{\max}=\log_2(K) $$

For the original 10-class problem:

$$ H_{\max}=\log_2(10)\approx3.3219 \text{ bits} $$

For the revised 38-class problem:

$$ H_{\max}=\log_2(38)\approx5.2479 \text{ bits} $$

The increase is approximately:

$$ 5.2479-3.3219=1.9260\text{ bits} $$

The larger class space creates greater maximum uncertainty when all classes are equally probable. However, the actual entropy depends on the new dataset's class distribution.

The number of output probabilities also increases from 10 to 38. A classifier must distinguish more classes, and similar diseases across different crops may create additional confusion.

For a uniformly random classifier, the expected accuracy under a balanced 38-class distribution is:

$$ \mathrm{Accuracy}=\frac{1}{38} \approx2.63\% $$

For a balanced 10-class problem, the corresponding baseline is 10%. These figures are theoretical reference points, not expected accuracies for a trained classifier. For imbalanced data, a majority-class predictor can achieve higher accuracy than the uniform random baseline while still performing poorly on minority classes.

A.2 Effect of Noisy Labels

Approximately 20% noisy labels means that a substantial fraction of training labels may not correspond to the true disease shown in the image.

This can cause a model to learn incorrect associations, increasing estimation error and potentially reducing its ability to generalise. A high-capacity model may memorise incorrect labels, while a strongly regularised model may be less able to fit them but may also underfit genuine disease patterns.

The noise level alone does not fully specify the problem. It matters whether incorrect labels are distributed uniformly or whether certain diseases are systematically confused with others.

A.3 Changed Computational Assumptions

The original analysis focuses on a desktop-style training and evaluation process. The revised requirements impose a strict inference latency of 50 ms per image and a 5 MB memory limit.

These constraints may rule out models that store many training images, maintain large ensembles or require expensive transformations during inference. The new system must be measured on the target hardware, with the preprocessing and prediction pipeline included in latency measurements.

A.4 Impact Summary

Requirement change	Consequence
10 to 38 classes	Larger output space and more opportunities for class confusion
Approximately 20% label noise	Greater risk of learning incorrect patterns
Maximum 50 ms per image	Requires low-latency inference
Maximum 5 MB model size	Requires careful control of model parameters and stored data

Smartphone deployment	Hardware-specific optimisation and memory testing become necessary

Interpretation: The original model-selection criteria are no longer sufficient. Predictive performance must be considered together with label-noise robustness, model footprint and smartphone latency.

B. Examination of the Original Conclusion [2 Marks]

B.1 Can the Original Best Model Meet the New Limits?

The original report has not yet established a best-performing baseline from measured experiments. Therefore, it is not possible to claim that Logistic Regression, k-NN, Extra Trees or a deep-learning model satisfies the 50 ms and 5 MB limits.

The following measurements are required for the best original baseline.

Measurement	**Required limit**	**Actual result**
Inference latency	At most 50 ms/image	To be measured
Model size	At most 5 MB	To be measured
Accuracy on 38 classes	No numerical target specified	To be measured
Macro F1-score	No numerical target specified	To be measured
Performance with noisy labels	Robustness must be evaluated	To be measured

Latency must be measured on the target smartphone or representative hardware. A model that meets the limit on a desktop may exceed it on a low-cost phone.

B.2 Sensitivity to 38 Classes and Noisy Labels

Logistic Regression will require additional output parameters for the larger class set, although the resulting model can still be compact.

k-NN can be sensitive to noisy training labels because incorrect labels may directly affect the majority vote among neighbouring examples. It also stores training examples, which can be a major memory burden under the 5 MB limit.

Extra Trees may capture nonlinear boundaries, but their model structures can consume more storage as the ensemble grows. They can also learn incorrect label patterns if the trees are sufficiently flexible.

These are algorithmic expectations. The actual sensitivity must be assessed using noisy-label experiments and the revised dataset.

B.3 Which Alternative Is Less Affected?

A compact, regularised parametric model is a reasonable candidate when strict memory and latency limits dominate. Unlike k-NN, it does not need to store the entire training dataset for prediction. Regularisation can also limit model complexity.

However, Logistic Regression's linear decision boundaries may not capture the complex visual differences among 38 crop–disease classes. A small quantised neural network may offer a better balance if it learns useful image features while satisfying the deployment limits.

Neither option can be declared the winner without measurements on the revised dataset and target hardware.

C. Analysis of Adaptation Options [2 Marks]

At least two adaptation strategies should be evaluated before selecting the final deployment design.

Option 1 — Change PCA Dimension and Regularisation

PCA can reduce the 4,096 raw features to a smaller representation, while regularisation limits the flexibility of Logistic Regression.

For example, 100 components retain 2.44% of the original feature count. Lower-dimensional inputs can reduce storage and computation, although the projection matrix also requires storage if it is part of the inference pipeline.

Increasing regularisation may reduce sensitivity to noisy labels, but excessive regularisation can cause underfitting. Reducing the PCA dimension may remove useful disease-discriminative features.

The PCA dimension and regularisation strength should be selected using validation performance on the revised dataset, not solely by explained variance.

Option 2 — Use Noise-Robust Training

Label smoothing replaces a one-hot target with a probability distribution that assigns a small amount of probability to other classes.

For a target class y, one common formulation is:

$$ y'_k= \begin{cases} 1-\epsilon+\epsilon/K,&k=y\\ \epsilon/K,&k\ne y \end{cases} $$

where $K=38$ and ϵ controls the smoothing amount.

This discourages extremely confident predictions and may improve generalisation. However, label smoothing does not identify which labels are actually incorrect and does not guarantee robustness to 20% noise. If the smoothing amount is excessive, the model may underfit.

Other strategies include noise-aware loss functions and careful identification or reweighting of suspicious training labels. These methods require validation to ensure that genuinely difficult disease examples are not incorrectly discarded.

Option 3 — Compress or Simplify the Model

Model compression can reduce memory and inference time.

Potential techniques include:

- Quantising model weights to lower-precision representations.
- Pruning redundant parameters or model structures.
- Using a compact CNN with fewer channels and layers.
- Applying knowledge distillation from a larger teacher model to a smaller student model.

For example, storing 1,000,000 parameters at 32 bits per parameter requires approximately 4 MB for the raw parameter values alone. At 8 bits per parameter, the same values require approximately 1 MB before accounting for metadata and other storage.

Thus, quantisation may help meet the 5 MB limit. However, quantisation and pruning can reduce predictive performance if important information is removed. Actual end-to-end memory and latency must be measured.

C.4 Comparative Summary

Adaptation	Potential benefit	Main risk	Evidence required

PCA and regularisation	Lower-dimensional input and controlled model complexity	Loss of useful features or underfitting	Macro F1, validation loss, latency and memory
Noise-robust training	Reduced sensitivity to unreliable labels	Excessive smoothing or incorrect noise assumptions	Clean-validation performance and noisy-label comparison
Model compression	Lower storage and inference cost	Accuracy degradation	Compressed model size, target-device latency and predictive metrics

Interpretation: These options address different problems. PCA targets input dimensionality, robust training targets label reliability, and compression targets deployment cost. A successful design may combine them, but the combined system must be evaluated rather than assuming that the individual benefits add together.

D. Evidence-Based Conclusion [2 Marks]

The original AgriVision design should be modified and re-evaluated rather than retained unchanged. The revised requirements introduce a larger class space, substantial label noise and strict smartphone deployment limits.

The mathematical analysis indicates that the maximum entropy increases when the number of equally probable classes increases from 10 to 38. The label-noise requirement introduces additional uncertainty into training, while the high-dimensional raw feature representation may increase computational costs. PCA and regularisation provide ways to control dimensionality and model complexity, but their effects on disease classification must be measured.

The experimental results needed to make the final decision are:

1. Accuracy, macro F1-score and per-class recall on the actual 38-class dataset.
2. A comparison between clean-label and noisy-label training.
3. The final compressed model size, including required inference components.
4. End-to-end inference latency on the target smartphone.
5. A comparison of the original best baseline with at least one adapted alternative.

If the original baseline exceeds either the 50 ms latency limit or the 5 MB model-size limit, it does not satisfy the revised deployment requirements. If it meets both limits but performs poorly under label noise or on minority classes, its predictive reliability remains insufficient.

A compact, regularised classifier is a sensible low-cost baseline, while a quantised compact CNN is a candidate if the classical model cannot adequately distinguish the 38 classes. The final choice should be the smallest and fastest model that meets the required predictive and operational criteria on the real dataset.

Final conclusion: The evidence currently supports modifying the original design and conducting targeted deployment experiments. It does not yet support claiming that any particular model meets all four requirements. The decision should be finalised only after the 38-class experiments and smartphone measurements are completed.